

Module 8: Electronic Data Exchange

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Electronic Data Interchange (EDI)

Electronic Data Interchange (EDI) is the electronic exchange of business documents and information between organizations in a standardized format without using paper documents or manual intervention. It enables computer-to-computer communication for business transactions such as purchase orders, invoices, shipping notices, bills, and payment information.

EDI is an important part of e-commerce because it automates business communication and improves coordination among trading partners. It helps organizations exchange accurate information quickly and securely.

Features of EDI

1. **Paperless Communication** – Business documents are exchanged electronically instead of paper documents.
2. **Standardized Data Format** – Uses common standards so different computer systems can understand the data.
3. **Computer-to-Computer Transfer** – Data moves directly between systems without manual typing.
4. **Fast Processing** – Transactions are completed quickly compared to traditional methods.
5. **Automation** – Reduces human involvement in document processing.
6. **Secure Data Exchange** – Information is transmitted through secure communication networks.
7. **High Accuracy** – Minimizes data entry mistakes and duplication errors.
8. **Integration with Business Systems** – EDI can connect with ERP, accounting, and inventory systems.

Benefits of EDI

1. **Faster Communication** – Documents are transferred quickly.
2. **Reduced Paperwork** – Minimizes paper usage and storage.
3. **Improved Accuracy** – Reduces manual data entry errors.
4. **Cost Reduction** – Saves administrative and processing costs.
5. **Better Inventory Management** – Provides real-time stock information.
6. **Increased Productivity** – Reduces repetitive manual work.

7. **Better Customer Service** – Faster processing improves satisfaction.
8. **Enhanced Security** – Secure systems protect business data.
9. **Strong Business Relationships** – Improves coordination between partners.
10. **Environmental Benefits** – Supports paperless operations.

Concepts of EDI

1. **Trading Partners** – Organizations exchanging electronic documents.

2. **Standardization** – Uses common formats for compatibility.
3. **Data Translation** – Converts data into EDI standard format.
4. **Electronic Transmission** – Transfers data through networks.
5. **Automation** – Reduces manual processing.
6. **Integration** – Connects with business systems.
7. **Document Exchange** – Transfers invoices, orders, and notices.

Applications of EDI

1. **Retail Industry** – Used for billing and inventory updates.
2. **Manufacturing** – Shares production and supply information.
3. **Banking and Finance** – Supports fund transfer and reporting.
4. **Transportation and Logistics** – Used for shipment tracking.
5. **Healthcare** – Exchanges medical and billing records.
6. **Government Services** – Used for tax and customs processing.
7. **E-Commerce** – Supports order and payment processing.
8. **Supply Chain Management** – Improves communication among partners.

Security in EDI

EDI Model

The EDI model explains the process through which electronic business documents are exchanged between organizations. **Steps in EDI Model**

1. **Document Preparation:** Business documents such as invoices or purchase orders are prepared electronically.
2. **Outbound Translation:** The sender's EDI software converts internal data into standard EDI format.
3. **Communication:** The formatted document is transmitted through the internet or Value-Added Network (VAN).
4. **Inbound Translation:** The receiver's EDI software converts the received EDI document into the organization's internal format.

1. **Authentication** – Verifies identity of trading partners.
2. **Encryption** – Protects confidential business information.
3. **Digital Signatures** – Ensures document authenticity and integrity.
4. **Access Control** – Restricts unauthorized access to EDI systems.
5. **Backup and Recovery** – Prevents data loss during system failure.

Limitations of EDI

1. High initial setup cost.
2. Requires technical expertise and training.
3. Compatibility issues may occur between systems.
4. Dependence on communication networks.
5. Small businesses may find implementation expensive.

Importance of EDI in E-Commerce

1. Enables fast electronic business transactions.
2. Supports paperless business operations.
3. Improves supply chain efficiency.
4. Reduces transaction processing time.
5. Enhances coordination among organizations.
6. Improves accuracy and operational efficiency.
7. Supports global trade and online business activities.

5. **Document Processing:** The receiving system automatically processes the document for further action.

Components of EDI System

- 1) **EDI Standards:** Rules and formats used for document exchange.
- 2) **EDI Software:** Converts business documents into EDI format and back into readable form.
- 3) **Communication Network:** Transfers EDI messages between organizations.
- 4) **Business Application:** Processes received information within company systems.
- 5) **Security Mechanism:** Provides authentication, encryption, and secure communication.

EDI Protocols

EDI protocols are standard formats used for exchanging electronic business documents between organizations. These standards ensure that different computer systems can communicate and understand business data correctly.

1) **UN/EDIFACT:** United Nations Electronic Data Interchange for Administration, Commerce and Transport) UN/EDIFACT is an international EDI standard developed by the United Nations for global electronic business communication. It is widely used in international trade, transport, banking, customs, and e-commerce systems. **Features of UN/EDIFACT**

1. **International Standard** – Accepted and used globally.
2. **Multi-Industry Support** – Used in trade, shipping, healthcare, banking, and government sectors.
3. **Standard Message Formats** – Provides common formats for invoices, orders, shipping notices, and payments.
4. **Supports Global Trade** – Helps businesses communicate across different countries.
5. **Improved Compatibility** – Different computer systems can exchange data easily.
6. **Electronic Business Transactions** – Enables fast and automated business communication.

Advantages of UN/EDIFACT

1. Reduces international trade paperwork.
2. Improves speed and accuracy of transactions.
3. Enhances coordination between global business partners.
4. Reduces communication and processing costs.

5. Supports secure electronic commerce operations.

Applications of UN/EDIFACT

- International trade
- Customs processing
- Shipping and logistics
- Banking transactions
- Government documentation

2) **ANSI X-12:** ANSI X-12 is an EDI standard developed by the American National Standards Institute (ANSI). It is mainly used in the United States and North America for electronic exchange of business documents. **Features of ANSI X-12**

1. **North American Standard** – Widely used in the USA and Canada.
2. **Standard Transaction Sets** – Provides formats for orders, invoices, payments, and shipping data.

3. **Industry Support** – Used in healthcare, retail, manufacturing, finance, and logistics.
4. **Reliable Communication** – Ensures accurate and secure data exchange.
5. **Business Automation** – Supports automated processing of transactions.
6. **System Integration** – Easily integrates with ERP and accounting systems.

Advantages of ANSI X-12

1. Improves speed of business communication.
2. Reduces manual errors and paperwork.
3. Increases operational efficiency.
4. Supports secure electronic transactions.
5. Enhances supply chain coordination.

Applications of ANSI X-12

- Retail business
- Healthcare systems
- Banking and finance
- Transportation and logistics
- Manufacturing industries

Basis	UN/EDIFACT	ANSI X-12
Developed By	United Nations	ANSI
Usage Area	International	Mainly North America
Scope	Global trade	US and Canadian business
Industries	Multiple international industries	Retail, healthcare, finance
Purpose	Global standardization	Regional business communication